FF-S-92B
August 27, 1974
SUPERSEDING
FF-S-92a
January 26, 1965

FEDERAL SPECIFICATION

SCREW, MACHINE. SLOTTED, CROSS-RECESSED OR HEXAGON HEAD

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers slotted, cross-recessed and hexagon head machine screws.

1.2 Classification.

1.2.1 Type and style. Machine screws shall be of the following types and styles (see 6.2).

Type I - Slotted (see fig. 1)

Style 1s	- Round head	Style 6s	- Oval head, 82°
Style 2s	- Flat head, 82°	Style 7s	- Truss head
Style 3s	- Flat head, 100°	Style 8s	- Binding head
Style 4s	- Fillister head	Style 9s	- Pan head
Style 5s	- Fillister head, drilled	Style 10s	- Hexagon head

Type II - Hexagon, Plain (see fig. 1)

Style 10p - Hexagon

Type III - Cross-Recessed, Recess Designs I and II (see fig. 2).

Style 1c	- Round head	Style 6c	- Oval head, 82°
Style 2c	- Flat head, 82°	Style 7c	- Truss head
Style 3c	- Flat head, 100°	Style 8c	- Binding head
Style 4c	- Fillister head	Style 9c	- Pan head

NOTE Use of cross-recess design II is not recommended.

FSC 5305

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

1.2.2 **Size.** Machine screws shall be classified according to size by the nominal diameter of the body.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications

QQ-A-200/3 - Aluminum Alloy Bar, Rod, Shapes, Tube and Wire, Extruded, 2024.
QQ-A-225/6 - Aluminum Alloy Bar, Rod and Wire, Rolled, Drawn or Cold Finished, 2024.
QQ-B-613 - Brass, Leaded and Non-Leaded, Flat Products (Plate, Bar, Sheet and Strip).
QQ-B-626 - Brass, Leaded and Non-Leaded Rod, Shapes, Forgings and Flat Products and Finished Edges (Bar and Strip).
QQ-B-637 - Brass, Naval Rod, Wire, Shapes, Forgings and Flat Products with Finished Edges (Bar, Flat Wire and Strip).
QQ-B-728 - Bronze, Manganese Rod, Shapes, Forgings and Flat Products (Flat Wire, Strip, Sheet, Bar and Plate).
QQ-B-750 - Bronze, Phosphor; Bar, Plate, Rod, Sheet, Strip, Flat Wire and Structural and Special Shaped Sections.
QQ-C-586 - Copper-Nickel-Zinc Alloy, Rod, Shapes and Flat Products with Finished Edges (Flat Wire, Strip and Bar).
QQ-C-591 - Copper-Silicon, Copper-Zinc-Silicon and Copper-Nickel-Silicon Alloys Rod, Wire, Shapes, Forgings and Flat Products (Flat Wire, Strip, Sheet, Bar and Plate).
QQ-N-281 - Nickel-Copper Alloy Bar, Plate, Rod, Sheet, Strip, Wire, Forgings and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought.
QQ-P-35 - Passivation Treatments For Corrosion-Resisting Steel.
QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-S-634 - Steel Bar, Carbon, Cold Finished (Standard Quality).
QQ-S-637 - Steel Bar, Carbon, Cold Finished (Standard Quality, Free Machining).
QQ-S-763 - Steel Bars, Wire, Shapes and Forgings, Corrosion-Resisting
QQ-W-321 - Wire, Copper Alloy.
QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements For
PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging and Packing For Shipment and Storage of.

Federal Standards.

Fed. Std. No. 66 - Steel, Chemical Composition and Hardenability.
Fed. Std. No. 123 - Marking For Domestic Shipment (Civil Agencies).
Fed. Test Method Std No 151 - Metals Test Methods

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale or a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications.

MIL-F-495 - Finish, Chemical, Black, For Copper Alloys
MIL-A-8625 - Anodic Coatings, For Aluminum and Aluminum Alloys.
MIL-C-13924 - Coating, Oxide, Black, For Ferrous Metals.
MIL-P-16232 - Phosphate Coatings, Heavy, Manganese or Zinc Base (For Ferrous Metals).
MIL-I-17214 - Indicator, Permeability, Low-MU (Go-No Go).
MIL-M-20693 - Molding Plastic, Polyamide (Nylon), Rigid.
MIL-B-24059 - Bronze, Nickel Aluminum Rod, Flat Products with Finished Edges, Shapes and Forgings.
MIL-C-81562 - Coatings, Cadmium, Tin-Cadmium and Zinc (Mechanically Deposited).

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1312 - Fasteners, Test Methods.
MS18116 - Bolt, Bolt-Stud, Stud, Stud-Bolt; Nickel-Copper-Aluminum Alloy, Special Requirements.
MS18211 - Screw, Machine - 80° and 100°, Flat Countersunk Head, Slotted, Plastic (Nylon).
MS18212 - Screw, Machine - Pan Head, Slotted, Plastic (Nylon), UNC-2A.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards Handbook

H28, Part I - Screw-Thread Standards For Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

American National Standards Institute (ANSI) Standard.

ANSI B18.6 3 - Machine Screws and Machine Screw Nuts.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway. New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 Materials. Unless otherwise specified (see 6.2), screws shall be made of the materials specified in 3.1.1 through 3.1.9.

3.1.1 Carbon steel. Carbon steel shall be in accordance with QQ-S-634, QQ-S-637 or Fed. Std. No. 66 Capped or rimmed steels shall be permitted. The minimum ultimate tensile strength shall be 60,000 PSI.

3.1.2 Corrosion-resisting steel. Corrosion-resisting steel shall be the 300 series in accordance with QQ-S-763 and Fed. Std. No. 66 or equal to or interchangeable with 16-18 or 18-8 chromium nickel alloy steel (developed for cold heading). The minimum ultimate tensile strength shall be 80,000 PSI. Straight chromium alloys of the 400 series shall not be used.

3.1.2.1 Magnetic permeability. Corrosion-resisting steel screws shall have a magnetic permeability of 2.0 maximum (air = 1.0) for a field strength of H = 200 oersteds using a magnetic indicator per MIL-I-17214.

3.1.3 Brass. Brass shall be in accordance with QQ-B-613, QQ-B-626 or QQ-W-321. The minimum ultimate tensile strength shall be 55,000 PSI.

3.1.4 Aluminum alloy. Aluminum alloy, 2024, shall be in accordance with QQ-A-200/3 or QQ-A-225/6 The minimum ultimate tensile strength shall be 62,000 PSI.

3.1.5 Copper alloys. Copper alloys shall be in accordance with QQ-B-637, QQ-B-728, QQ-B-750, QQ-C-586 or QQ-C-591. The mechanical properties shall be as specified in Table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

3.1.6 Nickel-copper alloy. Nickel-copper alloy shall be in accordance with QQ-N-281, Class A. The mechanical properties shall be as specified in Table II.

3.1.7 Nickel-copper-aluminum alloy. Nickel-copper aluminum alloy shall be in accordance with QQ-N-286, Class A. The mechanical properties shall be as specified in Table II.

3.1.8 Nickel-aluminum-bronze. Nickel-aluminum-bronze shall be in accordance with MIL-B-24059. The mechanical properties shall be as specified in Table II.

3.1.9 Plastic (nylon). Plastic (nylon) shall be in accordance with MIL-M-20693, composition A, type I. Plastic machine screws shall be as specified on MS18211 and MS18212.

3.2 Protective finish. Unless otherwise specified (see 6.2), screws shall be furnished uncoated with a naturally bright finish, not heat treated. When finishes are specified, they shall be in accordance with the following paragraphs.

3.2.1 Carbon steel screws.

3.2.1.1 Cadmium plating. Cadmium plating shall be in accordance with QQ-P-416, Type II, Class 3 or MIL-C-81562, Type II, Class 3.

3.2.1.2 Zinc coating. Zinc coating shall be in accordance with QQ-Z-325, Type II, Class 3 or MIL-C-81562, Type II, Class 3.

3.2.1.3 Phosphate coating. Phosphate coating shall be in accordance with MIL-P-16232, Type Z, Class 2.

3.2.1.4 Embrittlement relief. Electrodeposited carbon steel screws having a Rockwell Hardness of C40 or higher shall be subjected to an embrittlement relief treatment conducted in accordance with the applicable plating or coating specification.

3.2.2 Corrosion-resisting steel screws.

3.2.2.1 Passivation. Passivation treatment shall be in accordance with QQ-P-35.

3.2.2.2 Black oxide. Black oxide coating shall be in accordance with MIL-C-13924, Class 4.

3.2.3 Brass screws.

3.2.3.1 Black chemical. Black chemical finish shall be in accordance with MIL-F-495.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

3.2.4 Aluminum alloy screws.

3.2.4.1 Anodizing. Anodizing shall be in accordance with MIL-A-8625, Type I or II, Class 1.

3.3 Dimensions. Dimensions and tolerances shall be in accordance with the applicable Military Standard (MS) and ANSI B18.6.3.

3.4 Threads. Threads shall be Class 2A, UNC or UNF series, as specified (see 6.2), in accordance with Handbook H28. The method of forming shall be at the manufacturer's option

3.5 Workmanship. Screws shall be free from fins, burrs, sharp edges, cracks and surface contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all screws of the same material, type, style, size, thread and protective finish, produced under the same conditions and offered for acceptance at one time.

4.3 Sampling.

4.3.1 Sampling for examination. A random sample of screws shall be selected from each lot in accordance with MIL-STD-105, Inspection Level I. The Acceptable Quality Level (AQL) shall be as specified in Table I.

4.3.2 Sampling for tests. A random sample of screws shall be selected from each lot in accordance with MIL-STD-105, Inspection Level S-1. The AQL shall be 2.5 percent defective.

4.4 Examination Each screw taken as specified in 4.3.1 shall be examined to verify conformance with this specification. Examination shall be conducted in accordance with Table I. Any screw in the sample containing one or more defects shall be rejected, and if the number of defective screws in any category exceeds the acceptance number for that category, the lot represented by the sample shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of defects

Category	Defect	Inspection Method
Critical	None defined	
Major	AQL = 2.5 percent defective	
101	Size (1.2.2)	SIE*
102	Length (3.3)	SIE
103	Thread (3.4)	SIE
104	Slot or recess dimensions (3.3)	SIE
105	Bearing surface of flat head screws (3.3)	SIE
Minor	AQL = 4.0 percent defective	
201	Magnetic permeability (3.1.2.1)	SIE
202	Other dimensions (3.3)	SIE
203	Protective finish missing or incomplete (3.2)	Visual
204	Workmanship (3.5)	Visual

* SIE = Standard Inspection Equipment

4.5 Tests.

4.5.1 Tensile strength. Each screw taken as specified in 4.3.2 shall be tensile strength tested in accordance with Test No. 8 of MIL-STD-1312 to verify conformance with 3.1.

4.5.1.1 Screws too short to be tensile strength tested shall be hardness tested in accordance with Test No. 6 of MIL-STD-1312. Screws shall meet the minimum hardness requirement of the material from which they were made.

4.5.2 Bending. When ferrous screws are of sufficient length in the unthreaded portion to permit cold bending, each screw taken as specified in 4.3.2 shall withstand being bent 160° without fracture, to a curve having an inner radius equal to the major body diameter of the screw.

4.5.2.1 When the bending test cannot be applied, each test sample shall withstand flattening cold to a thickness equal to one-half of its body diameter, without cracking.

4.6 Protective finish. Examination and test of protective finisnes shall be in accordance with the applicable specification of 3.2.

4.7 Preparation for delivery. Examination and test of preparation for delivery shall be in accordance with PPP-H-1581, to verify conformance with Section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

4.8 Chemical analysis. The supplier shall furnish a mill certificate specifying the chemical composition of the material used in the manufacture of the screws. When specified (see 6.2), chemical analysis shall be in accordance with Method 111.2 or 112.2 of Fed. Test Method Std. No. 151. In case of dispute, Method 112.2 shall apply.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Packaging shall be Level A or C and packing shall be Level A, B or C in accordance with PPP-H-1581, as specified (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to marking required by the contract or order, all interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to marking required by the contract or order, all interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Machine screws are intended to be used in the assembling of parts in relation to each other, either permanently or temporarily.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, style and size (1.2).
(c) Material (3.1).
(d) Protective finish, when required (3.2).
(e) Thread series (3.4).
(f) Chemical analysis, when required (4.8).
(g) Selection of applicable level of packaging and packing (5.1).
(h) Applicable MS part number.

6.3 Military procurement. Items procured under this specification for military use are to be limited, wherever possible, to the variety shown on the applicable Military Standard (MS).

6.4 Preferred screws. It is recommended that pan head screws be used in lieu of round, truss or binding head screws and that cross-recessed drive (design I) be used in lieu of slotted drive.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

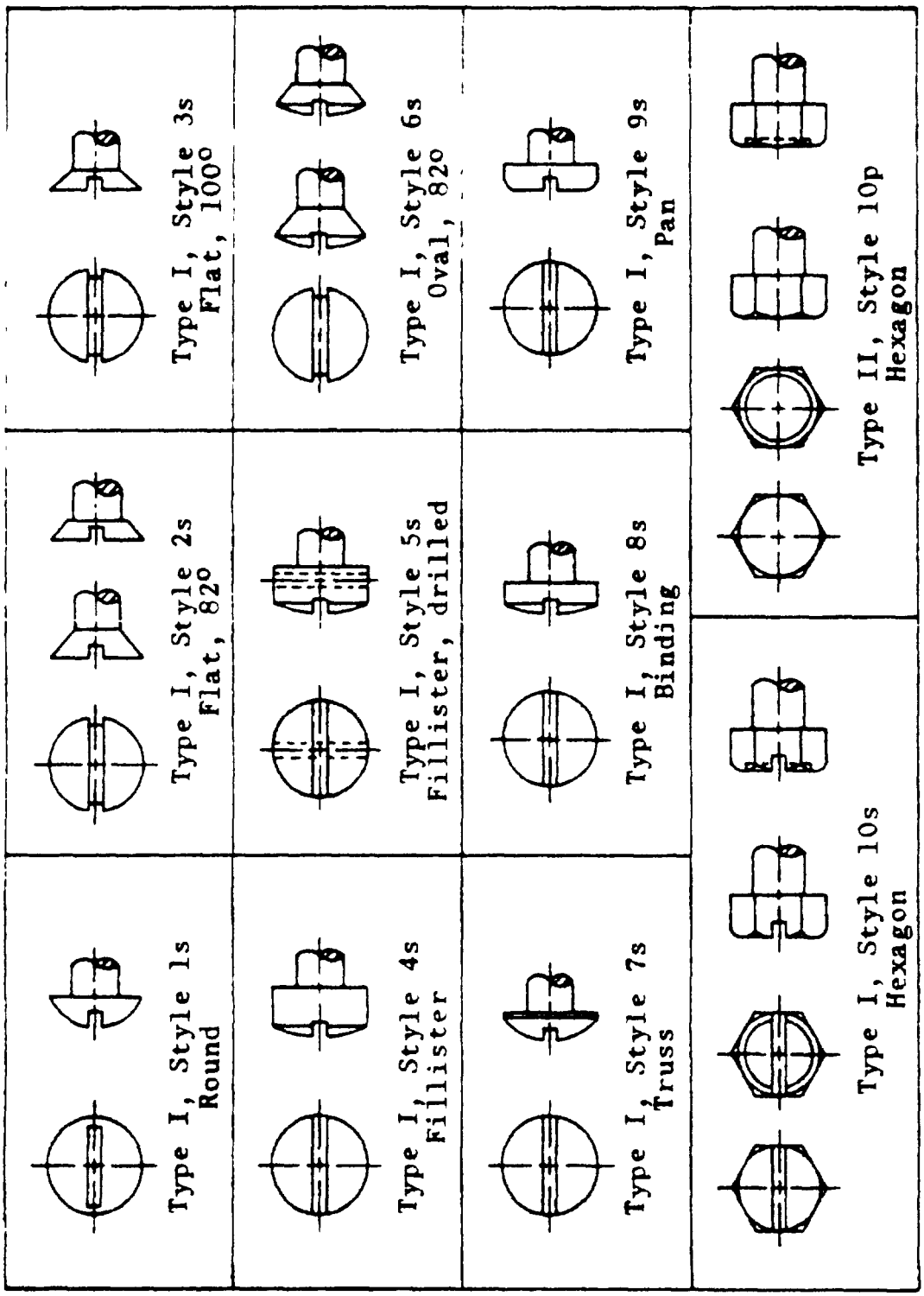
TABLE II. Mechanical properties of nonferrous machine screws

Material	Applicable Document	Composition or class	Condition	Ultimate Tensile Strength psi. Min	Yield Strength psi. Min	Elongation 1/ (min.) Percent
Copper-nickel-zinc	QQ-C-586	Optional	Quarter-hard	58,000	----	Meets bend test
Manganese bronze	QQ-B-728	Class A	Soft	55,000	22,000 3/	20
Nickel-aluminum-bronze	MIL-B-24059	----	Extruded and annealed	90,000	50,000 3/	18
Phosphor bronze	QQ-B-750	Comp. A	Hard	60,000	35,000 2/	15
Silicon bronze	QQ-C-591	Copper alloy No. 651	Hard	60,000	40,000 3/	10
Naval brass	QQ-B-637	Copper alloy No. 462 or 464	----	60,000	27,000 3/	22
Nickel-copper alloy	QQ-N-281	Class A	----	80,000	40,000 2/	20
Nickel-copper-aluminum alloy	QQ-N-286 MS18116	Class A	Annealed and age hardened	130,000	90,000 2/	20

1/ In 2-inch gage length.
2/ 0.2 percent offset.
3/ 0.5 percent extension under load.

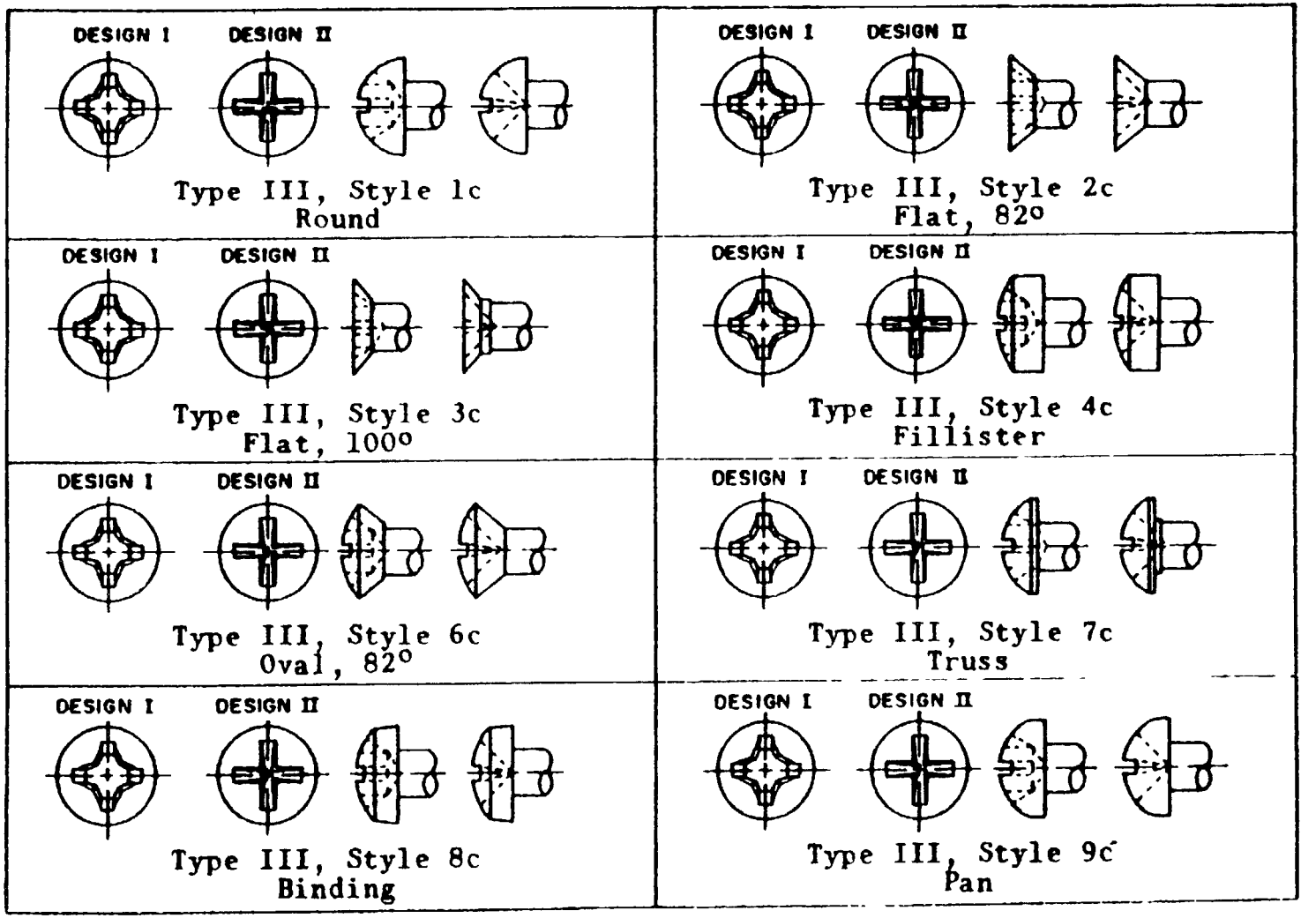
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

FIGURE 1: Type I - Slotted and Type II - Hexagon, Plain



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

FIGURE 2: Type III - Cross Recessed



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

Military Custodians
Army - WC
Navy - None
Air Force - 82

Reviewer Activities:
Army - AV, EA, MU
Navy - None
Air Force - None
DSA - IS

User Activities:
Army - AT, CE, ME, PA, WT
Navy - AS, MC, OS, SH, YD
Air Force - None

Preparing Activity:
Army - WC

Project No. 5305-1221

Civil Agencies Interest.
GSA-FSS
COM-NBS

U. S. GOVERNMENT PRINTING OFFICE : 1974 - 585-644/1353

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 35 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armement Research and Development Command
ATTN: DRDAR-TST-S
Dover, NJ 07801

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.